

NAME

help -- print online documentation for the named modules or packages

USAGE

help [template]

PARAMETERS**template**

A string listing the modules or packages for which help is desired. Each list element may be a simple name or a pattern matching template. Abbreviations are permitted. If *template* is omitted a menu will be printed for the current package, listing each task (or subpackage) and describing briefly what it is.

file_template= no

If this switch is set the template is interpreted as a template, and all help blocks found in the named files are output. The help database is not searched, hence manual pages can be printed or documents may be formatted without entering the files into a help database. In other words, "help file.hlp fi+" makes it possible to use *help* as a conventional text formatter.

all = no

Print help for all help modules matching *template*, rather than only the first one found.

parameter = "all"

If the value of this parameter is not "all", only the help text for the given parameter will be printed.

section = "all"

If the value of this parameter is not "all", only the help text for the given section (e.g. "usage", "description", "examples") will be printed.

option = help

The option parameter specifies the type of help desired, chosen from the following:

help Print the full help block for the named module.

source

Print the source code for the module (which often contains additional detailed comments).

sysdoc

Print the technical system documentation for the named module.

directory

Print a directory of all help blocks available for the named package.

alldoc

Print all help blocks in the file containing the help block for the named procedure (i.e., both the user and system documentation).

files

Print the names of all help files associated with the named modules or packages.

summary

Print only the titles and sizes of help blocks in referenced help files. The contents of the blocks are skipped. Titles are printed for *all* help blocks found in the file containing the help block for the named module.

page = yes

Pause after every page of output text. Turning this off for large documents speeds up output

considerably.

nlpp = 59

The number of lines per page if output is redirected, e.g., to *lprint*.

lmargin = 1

Left margin on output.

rmargin = 72

Right margin on output.

search = no

If enabled the template is interpreted as a search string and the task is started with the search panel open with the results of the search. The *file_template* parameter is ignored with search turned on.

home = ""

The home page for the task. If not set and no template is specified the task will start with the online help in the main window, otherwise it may be set to a filename to be displayed when the task starts. This file may contain a text help block which will be formatted before display, or it may be a valid HTML file. See below for a description of the format of a homepage file which provides links to tasks.

printer = "printer"

Default hardcopy printer name. If the *value* of the parameter is reserved string "printer", the actual device is the value of the CL environment variable *printer*.

showtype = no

Add task-type suffix in package menus?

quickref = "uparm\$quick.ref"

Name of the quick-reference file used for searching. This file is the first time the task is run in GUI mode or whenever it doesn't exist, or when any help database file has been updated.

uifname = "lib\$scr/help.gui"

The user interface file. This file is what defines the look and behavior of all the graphical user interface elements. Experts may create variants of this file.

helpdb = "helpdb"

The filename of the help database to be searched. If the *value* of the parameter is the reserved string "helpdb", the actual filename is the value of the CL environment variable *helpdb*.

device = "terminal"

Output device if the standard output is not redirected. Allowable values include:

terminal

If the *value* of the parameter is the reserved string "terminal", the actual device name is the value of the CL environment variable *terminal*.

text Output the formatted help page as plain text.

gui Invoke the GUI for browsing the help system. This option will only work if the *stdgraph* environment variable is set the *xgterm*, and the user is running IRAF from an *XGterm* window.

html Output the formatted help page as HTML text.

ps (or postscript)

Output the formatted help page as postscript.

BASIC USAGE

Despite the complex appearing hidden parameters, **help** is easy to use for simple tasks. **Help** is most commonly used to get help on the current package, and to get help on a program named in a the current package one need only type **help** without any arguments. For example, if the current package is **plot**, the command and its output might appear as follows:

```
pl> help
      contour - Make a contour plot of an image
      graph   - Graph one or more image sections or lists
      pcol    - Plot a column of an image
      pcols   - Plot the average of a range of image columns
      prow    - Plot a line (row) of an image
      pros    - Plot the average of a range of image lines
      surface - Make a surface plot of an image
pl>
```

To get help on a module one supplies the module name as an argument,

```
pl> help graph
```

and the manual page for the **plot.graph** program will be printed on the terminal. To get a hardcopy of the manual page on the printer, the output may be redirected to the line printer, as follows:

```
pl> help graph | lprint
```

DESCRIPTION

The function of the **help** program is to perform a of the help database *helpdb*, printing help for all packages and modules matching the template. By default the standard IRAF help database is but any other help database may be searched if desired. A help database is precompiled with the **mkhelpdb** program to speed up runtime searches for help modules. The standard IRAF help database contains the documentation and source for all CL programs and system and math library procedures installed in IRAF.

A help template is a string type parameter to the CL. The form of a is a list of patterns delimited by commas, i.e.,

```
"pattern1, pattern2, ..., patternN"
```

The form of a pattern is

```
package_pattern.module_pattern
```

If the "." is omitted *module_pattern* is assumed. The standard pattern matching meta-characters, i.e., "***?[]**", are permitted in patterns. Simple patterns are assumed to be abbreviations.

GUI OPERATION

The GUI component of the task is a front-end to the IRAF **help** task which provides on-the-fly conversion of help documents to HTML for presentation in the GUI or formatted PostScript for hardcopy. The GUI is started by setting the *device* parameter to the special value *gui*, it is only available when using an XGterm window to start IRAF and assuming the *stdgraph* environment variable is set to *xgterm*.

Help pages may be loaded on the command line, through use of a file browser, or by navigating the help databases using a familiar CL package menu scheme. It also features a search capability similar to the **references** task and a complete history mechanism.

When invoked with no command line arguments the task starts as a `task` and the user is presented with a GUI that has the toplevel CL package menu in the upper navigation window. The main display window below will contain any help page specified in the template parameter or loaded on the command line by specifying the template and file_template parameters. If the search parameter is enabled the template is taken to be a search phrase and the database is searched for tasks matching the keyword and the GUI will appear with the search panel mapped so the user can select the task help to view. When no template is given the main display window will start with the page specified by the home parameter, this can be a user-defined HTML file giving links to specific tasks (see below for details) or if home is empty the display will contain the online help for the task.

The first time the task is run, or whenever the help database is updated, a quick reference file (specified by the task quickref parameter) and package menu file will be created in the user's *uparm* directory to speed up help searching and subsequent startups of the task.

NAVIGATING THE HELP SYSTEM

When run as a GUI browser *HELP* works very much like any WWW browser. The top panel is a list widget that will always contain a CL package listing, at startup this will be the toplevel "Home" package menu one would see when first logging into the CL containing the core system packages, NOAO package, and any site-specific external package, or in the case of starting with a specific task it will be the parent package for the task. Additionally, system documents for the **os** HSI routines and the **imfort** and **math** interfaces will be available in the *Home* package although these are programmatic interfaces and not tasks which can be executed.

New packages or task help pages are loaded by selecting an item from the package menu list using the left mouse button. If the requested item is a package, the menu listing will change as though the package were loaded in the CL, and the help display panel will contain a list of tasks with a one-line description for each task such as would be seen with a "help <package>" command using the standard task. New items may then be selected using either the menu list or links in the display panel. If the item is a task, the help page for the task will appear in the display panel. In either case new pages may be selected from the menu listing.

Specific help documents may also be requested by entering the task/package name in the **Topic** text widget above the menu list. As when selecting from the package menu list, items selected this way will cause the menu list to change to the package menu for the parent package if task (displaying the help page in the display panel) or the package menu if the item is a package (displaying the one-liner package listing in the display panel).

Using the **Back** button will revert to the previous page in the history list which will either be the previously loaded package or help page. Similarly, selecting the **Forward** button will move the next page further down in the history list, either button will become insensitive when the end of the list on either end is reached. Selecting the **Up** button will cause the browser to immediately jump up the previous package, skipping over any help pages that were loaded in between. The **Home** button will cause the default homepage (either the user-defined page if specified by the task *home* parameter or the online help) to be displayed. Browsing in this way can also be done using the navigation menu created by hitting the right mouse button while in the main display panel.

Users can also jump to specific pages in the history list using the **History** button on the main menubar. The right column of the menu will indicate whether the item is a task, package, internal link or a file. The history list is truncated at about 40 entries in the menu but the user may work back incrementally by selecting the last item of the menu, after which the History button will display the previous 40 entries. The history list may be cleared except for the current page by selecting the *Clear History* menu item.

BROWSING A HELP DOCUMENT

Once a help page is loaded the middle menubar above the display panel will change to activate widgets based on the position within the history list and options available for a particular page. The left-most group of buttons are the standard navigation buttons described above. The middle group of buttons contains the **Sections** and **Parameters** buttons which are used to browse within a . The *Sections* button is a menu listing all of the sections found within a help page, allowing the user to jump to a rather than scrolling through the entire document. The *Sections* menu is also available using the middle mouse button from the main display area. The *Parameters* button is similarly a menu listing of all task parameter help sections found within the document. Both or either of these buttons will become insensitive when no section or parameter information is found in the document.

The right-most group of buttons represent the various help options available for each page. The default is to get the task help, however help pages may have an associated **source** file or **sysdoc** (e.g. if a CL script there may be a pointer to the script source itself, or a package may have a general overview document listed as the system document). Once a help page is loaded these buttons will change become sensitive if is available, simply select the button to view the option. Selecting the **Files** button will bring up a panel listing all the files associated with a particular help topic. When a help topic is selected and an option is defined but the file does not exist, the options button will display a yellow diamond icon even if the button is insensitive, a green icon indicates the currently selected option. This feature may be disabled by selecting the "Show missing files" item from the main menubar **Options** menu.

SEARCHING

Searching the help database is done by selecting the **Search** button from the main menubar to bring up the search panel. Users may then enter one or more keywords into the **Topic** field at the bottom of the panel and initiate the search with either a carriage return or hitting the *Search* button just beside it. The panel will then show a list of all tasks and packages which match the search phrase along with a description of the task. Help pages may be displayed by selecting either the task or package link with the left mouse button, in both case the package menu list on the main help window will be updated to list the package contents allowing other tasks from that package to be selected in the normal way.

By default the exact phrase entered in the topic window will be used for the search. This can be relaxed by toggling the "Require exact match" button at the top of the panel. For example, to search for all tasks matching *either* the keyword "flat" or "field" turn off the exact match toggle and the search will return not only tasks matching "flat field" but also any task description containing only one of the words such as the VELVECT task which plots velocity *fields*.

Within a help document itself one can search for a the **Find** button from the main menubar to bring up a enter the search string. When the text is entered the main display window will reposition itself and highlight the text found within the document. Searches can be repeated and will wrap around the document automatically, searches can be done either forward or backward through the text and may be case insensitive.

USER_DEFINED HOME PAGES

By default the *help* GUI will start with the online help page displayed in the main help window. The user can change this by setting the task **home** parameter to be a path to any valid file. This file may be plain text, a help document in LROFF format which will be converted to HTML for display, or a native HTML document.

HTML files may contain URLs of the form

```
<a href=[package.]task>url_text</a>
```

where *url_text* is the text to appear in the window and the URL itself consists of an optional package and task name delimited by a period. For example, to create a link to the **splot** task in a document one would

use the URL

```
<a href=onedspec.splot>splot</a>
```

In this way users can create a homepage which serves as a file or index of shortcuts to the most commonly accessed help pages.

LOADING FILES

Text files may be loaded on the command line when starting the task by specifying the filename and setting the `file_template` task parameter. The named file will be searched for a `.help` LROFF directing indicating it a help block that will be converted to HTML for display. If no help block is found the file will be displayed as-is, meaning existing HTML documents can be loaded and will be formatted correctly.

Once the task is running users may load a file by selecting the **Open File...** menu item from the main menubar **File** menu or the right-mouse-button menu from within the main display area. This will open a file browser allowing users to change directories by using the navigation buttons at the top of the panel, or selecting items from the leftmost directory listing. Selecting a file on the rightmost list will cause it to be loaded and automatically formatted if it contains a help block. The file list may be filtered to select only those files matching a particular template by changing the **Filter** box at the top of the panel. Filenames or directories may be entered directly using the **Selection** box at the bottom of the panel.

SAVING FILES

Once a file has been loaded in the browser it may be saved to disk as either *source* (i.e. the original LROFF file if that was converted for the display, or whatever file is currently displayed regardless of format), *text* to save formatted plain text such as that produced by the standard **help** task, *HTML* to save the converted HTML used in the display, or *PostScript* to save formatted PostScript of the document such as that sent to the printer using the **Print** button. Not all options will be available depending on the format of the input text, unavailable options will be insensitive in the GUI.

The **Save** panel is opened by selecting the **Save As...** menu item from the main menubar **File** menu or the right-mouse-button menu from within the main display area. The file browser operates the same as when loading images, the only difference is that file selection simply defines the filename to be used and does not cause the save to occur automatically. Users can overwrite existing files by selecting the *Options* toggle at the bottom of the panel.

HARDCOPY OUTPUT AND SAVING DISK FILES.

Help pages may be output to any configured IRAF printer by selecting the main menubar **Print** button to bring up the print panel. Task help pages will be converted to formatted PostScript and may be sent to either a printer or saved to disk depending on the selection made in the printer panel. If the printer name is set to the special value "printer" then the device named by the CL *printer* environment variable will be used. When saving to disk files the default action is to save to a name is the task name plus a ".ps" extension. Either of these are changeable within the GUI as is the default page size to be used when generating the PostScript.

The main menubar **File** button can also be used to bring up the file browser in order to save the current document to disk. Help pages may be saved as either the origin LROFF source for the file, formatted text as you would get from the standard help task, HTML as is displayed in the GUI, or formatted PostScript. The choice of formats is dictated by the type of file being displayed (e.g. you cannot save PostScript of a

LROFF DIRECTIVE EXTENSIONS FOR HTML

To better support HTML links within documents and to other help pages two new directives have been added to the LROFF text formatter. These are **.hr** to specify a link (an HTML *HREF* directive) and **.hn** to specify a name (an HTML *NAME* directive). The syntax for these are as follows:

```
.hn <name>
.hr <link> <text>
```

where *<name>* is the destination name of an internal link, *<link>* is the URL of the link to be created, and *<text>* is the text to be displayed in the HTML. The URL syntax is either a by a destination name, a simple *task* name or *package* name, or a *package.task* pair giving a more precise task. For internal links the current document is repositioned so the name is at the top of the display, for task help links new help pages will be loaded in the browser.

These directives are ignored when converting the LROFF to either formatted plain text or PostScript.

GUI EXAMPLES

- 1) Start *help* as a GUI browser:

```
cl> help dev=gui
```

- 2) Begin by searching for the phrase 'gauss', tasks and packages may be selected from the search panel which will appear when the task starts:

```
cl> help gauss dev=gui search+
```

- 3) Load an LROFF help page in the browser at startup

```
cl> help mytask.hlp dev=gui file+
```

EXAMPLES

1. Print the help text for the program *delete* in the package *system* (output will be directed to the terminal):

```
cl> help system.delete
or
cl> help delete
or
cl> help del
```

2. Print the help text on the line printer:

```
cl> help delete | lprint
```

3. Print help for the current package:

```
cl> help
```

4. Print the usage section of all modules in the package **images**:

```
cl> help images.* section=usage
```

5. Print a directory of all help blocks in the packages **clpackage** and **clio** (and any others whose names begin with the string "cl"):

```
cl> help cl* op=dir
```

6. Print a directory of each package in the database (useful for getting an overview of the contents of a help database):

```
cl> help * op=dir
```

7. Print the source for all of the string utilities in the system library package **fmtio**:

```
cl> help fmtio.str* op=source
```

8. Find all tasks that delete something:

```
cl> help * | match delete
```

9. Print the manual pages for the *help* and *lprint* tasks on the default printer device:

```
cl> help help, lprint | lpr
```

10. Capture the manual page for task *hedit* in a suitable for printing on any device.

```
cl> help hedit dev=text > hedit.txt
```

11. Print the manual page for task *hedit* as a

```
cl> help hedit dev=ps | lprint
```

BUGS

On some systems, typing the next command keystroke before the end-of-page prompt is printed may result in the character being echoed (messing up the output) and then ignored when raw mode is enabled for the prompt.

SEE ALSO

references , phelp , mkhelpdb , hdbexamine , mkmanpage , lroff , the online task help documents.